UNILAK RWAMAGANA CAMPUS

SOFTWARE ENGINEERING AND USER CENTRED DESIGN

Citizen Issue Reporting and Escalation System

CIRES

Milestone 1

Problem Discovery and User Research

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Problem discovery and user research

We chose road and drainage reporting because we wanted to understand what happens after a resident contacts a local office. We are working on CIRES, the Citizen Issue Reporting and Escalation System. We prepared the interview questions together, mapped who handles each step and used a blocked drain example to discuss where a report might get delayed or lost.

Five tests for our problem

We will check the problem against the five tests in slide 17. A recent incident will help us check whether the problem is real. We will compare accounts to see whether it happens more than once and record any calls, visits or waiting time people describe. We can ask a resident, an officer and a supervisor about the same stages. We kept the proposed build to one area, two report categories and one escalation step so the team can complete a pilot within the course.

Five questions before we design

How did the resident report the last road or drainage issue?

What happened after the office received the report?

How did staff decide who should act and by when?

How did anyone notice that a report was overdue?

How did the resident find out what had been done?

We will use these questions to check whether the difficulty is in reporting the issue, passing it to the right officer or finding out what happened afterwards.

Users at each stage of the project

We start by asking residents and staff how they handle reports today. We then turn the needs they describe into requirements. During design, we will ask users to try the reporting form and case list. During testing, we will ask a resident to find an update and a supervisor to find an overdue report. Their feedback will guide changes before release.

Who uses, pays, approves or is affected

We considered who reports the issue, who works on it and who can act when it is overdue. The resident and the staff handling the report are the main users. We also included receiving staff and maintenance workers because they can explain how a description reaches the person doing the work.

Stakeholder	Role	Power	Interest
Resident who submits a report	Reports the issue and needs a reference and progress updates	Low	High
Assigned officer	Accepts the case, records work and explains the outcome	High	High
Supervisor	Assigns reports and follows up overdue cases	High	High
Local office receiving staff	Receive the report and decide whether the description is usable	High	High
Maintenance personnel	Carry out the work and provide action information	Low	High

Pilot-area leadership	Set authority and operating rules	High	Low
Residents outside the pilot area	Part of the wider service context	Low	Low

Table 2. Our stakeholder power and interest map

Power	**Lower interest**	**Higher interest**
High	Keep satisfied. Pilot-area leadership.	Involve closely. Supervisor, assigned officer and local office receiving staff.
Low	Monitor. Residents outside the pilot area.	Keep informed. Residents who submit reports and maintenance personnel.

We plan to involve the officer, supervisor and receiving staff closely. We will keep residents who report issues and maintenance workers informed. Pilot-area leadership sets the local working rules, and we will monitor whether the pilot affects residents outside the area.

We grouped the wider stakeholders by their role. Residents, frontline officers and supervisors use the pilot directly. Receiving staff and maintenance workers help us understand handoffs. For a wider rollout, local government or another project sponsor could fund staff time, hosting and support. We have yet to confirm a funder. Pilot leadership would need to agree who can assign cases and set deadlines. Residents with limited internet access, including people who need help using a form, may be affected even if they never sign in.

Role in a wider rollout	**What we would confirm**
Pays	Who would cover staff time, hosting and support. The pilot has no confirmed sponsor.
Approves	Who can authorise the pilot boundary, response deadlines and access for officers.
Supports	Who would manage accounts, help residents report issues and maintain the service.
Affected	How residents without reliable internet, including people who need assisted reporting, would reach the local office.

Existing reporting routes and related services

A resident can explain a problem during an office visit, call or send a message. An office can use a paper register to keep a record. What we want to check locally is how a resident or officer finds the same entry again and how a later action is linked to it.

Service	**What we can learn from it**

FixMyStreet	Residents report local issues by location and the service routes reports to the relevant authority. We will compare its locationbased reporting with our plan for a report reference and local officer assignment.
SeeClickFix 311 CRM	Staff can manage, assign and update requests. We used this example to think about the officer queue and the updates a resident can see.
MINALOC citizen complaint services	The service offers ways to submit complaints and forward them to the responsible institution. We want to test a narrower road and drainage workflow with a due time for each assigned report.

We used provider descriptions for this comparison. We still need to check which routes are used in the pilot area.

Ask what happened last time

We will ask each person to describe a recent incident in the order it happened. We will ask for permission to take notes and use participant codes. For each account, we will record how the issue was reported, what the first response was, who received it next, what the participant did to follow up and how it ended.

P1 Resident

Tell us about the last road or drainage issue you reported. How did you know the office received it? What did you do to get an update? What happened next? How long did it take? If you are comfortable showing us the record or message, what did it say?

Interview record	**Details from the participant**
Date, place and consent	28th Oct, MUHAZI , RWAMAGANA
Channel and issue category	ROAD DRAINAGE
What happened and who handled it	Went to see cell leader , for solving it
Follow-up and outcome	It took time to be fixed
Exact quote	The service was poor

P2 Officer or receiving staff

Tell us about the last report you handled. How did you know who should work on it? What did you record when the work ended? What happened next? How long did it take? If you are comfortable showing us the record or message, what did it say?

Interview record	Details from the participant
Date, place and consent	29th Oct MUHAZI Cell
Channel and issue category	Farm conflict
What happened and who handled it	Sector land office
Follow-up and outcome	It took 1 months to be fixed
Exact quote	Need improvement in QoS

P3 Supervisor

Tell us about the last report that passed its deadline. How did you find out? What information did you use to follow it up? What happened next? How long did it take? If you are comfortable showing us the record or message, what did it say?

Interview record	Details from the participant
Date, place and consent	30th Oct MUHAZI
Channel and issue category	Road drainage report
What happened and who handled it	Executive director
Follow-up and outcome	done
Exact quote	It will be good if there's a way for centralized request

Grouping the problems we need to check

We grouped five possible problems by the part of the report they affect. The table shows what would help us confirm each concern in the field.

Concern	What we will look for
PP1 Receipt	A resident's account of what proof or reference they received.
PP2 Ownership	An account or record showing who received the case and how its deadline was set.
PP3 Progress	An example of how the resident checked what was happening.
PP4 Delay	An account of how a supervisor noticed a missed deadline.
PP5 Closure	A record or message showing what the resident was told when work ended.



Figure 1. Our affinity map of possible reporting problems

Who we are designing for

Our first user profile is a resident who needs to report a road or drainage issue and find the same case later. The resident needs a reference and a short update that makes the next step clear. Our second profile is a local staff member who needs the location, the person assigned and the due time. The officer uses that information to act, and the supervisor uses it to follow up overdue reports. These are working profiles based on the walkthroughs.

Our problem statement

Residents in our proposed pilot area who report a damaged road or blocked drain may struggle to find out whether the office received the report, who is handling it and what has been done. If the office and the resident cannot refer to the same record, the resident may call again and repeat the location. An overdue report may also escape the supervisor’s attention. Our proposal and walkthroughs helped us identify this possible problem. We still need interview notes and any office observations to confirm its cause and effect.

Our work in the discovery stage

We plan to give each report a reference. The supervisor will assign it to an officer and set a due time. The resident will see whether the case was received, assigned, in progress or resolved. If the case stays open after its deadline, it will appear once in the supervisor’s overdue queue. The officer will record what was done before closing it.

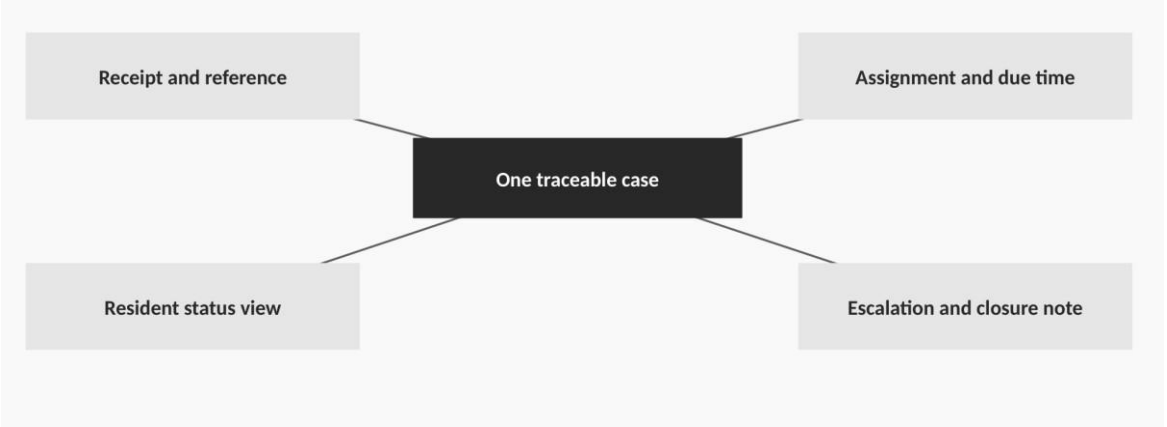


Figure 2. Ideas we considered for keeping the report traceable

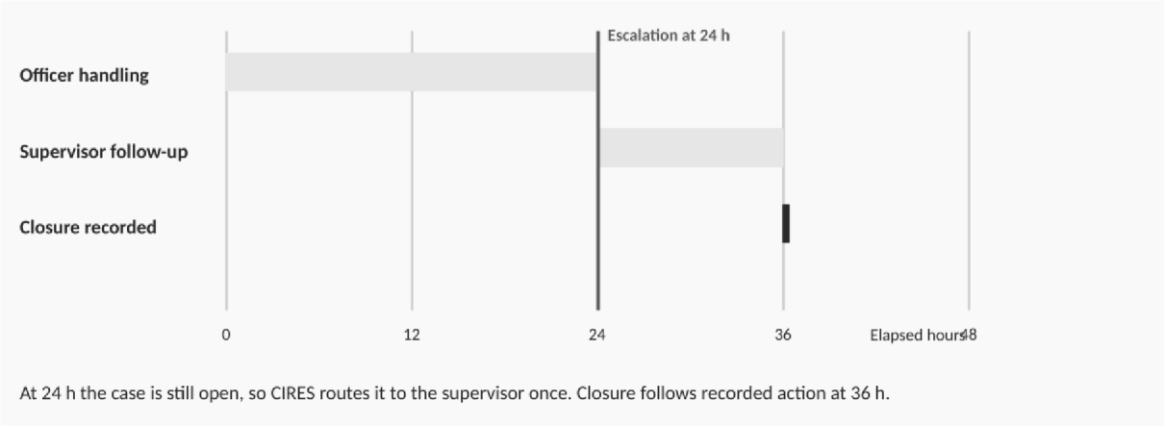


Figure 3. Example of an overdue report and its closure

In this example, the report is flagged at 24 hours and closed at 36 hours. These times illustrate the sequence. We will agree the pilot deadline with the people responsible for handling reports.

We used the stakeholder and affinity maps to organise the people and problems we need to study. We will screen for people who recently reported, handled or followed up a road or drainage issue. We will record what they did before asking what they would change.

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